SHARDA UNIVERSITY — SCHOOL OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING | CSCR2600 PBL-1

Final Project Implementation & Evaluation Report: Expenzo

Full-Stack Implementation, 184-Test Automated Verification, Defect Log, UAT Protocol, and Demo Script

Evaluation Milestones: Implementation Review (#R2) & Result Review (#R3) | **Session:** 2026–2027
Authors: Aryan Kaundilya (Full-Stack Architecture & Backend), Devanshi Singh (Frontend UI/UX & QA)
Production URL: https://expenzo-red.vercel.app/ | **Test Battery:** 184 Tests Passed (100%)

1. Executive Summary & Production Scope

Expenzo is an enterprise-grade personal finance intelligence platform deployed on Vercel Edge with a hosted Supabase PostgreSQL backend. It features 17 normalized tables, kernel-level Row Level Security, atomic stored procedures, and a hybrid AI co-pilot. All features are verified via 184 automated tests across 20 suites.

2. Module Coding & Architecture (Rubric R2 - PO3)

The software comprises nine decoupled modules in React 19 and TypeScript 5.8: Authentication, Financial Accounts, Canonical Ledger, Atomic Transfer Engine, Budget Envelopes, Recurring Obligations, Goal Tracking, Hybrid AI Co-Pilot, and Analytics Reporting. Global state is managed via Zustand.

3. Automated Verification Battery (Rubric R3 - PO6)

The test battery was executed using Vitest 3.0.5, achieving a 100% pass rate across 184 tests in 20 suites:

Test Suite	Module Under Test	Tests	Pass Rate	Verification Focus
authStore.test.ts	Authentication	9	100%	Session tokens, auto-cleanup
currency.test.ts	Integer Math	12	100%	Paise precision, zero drift
accounts.test.tsx	Accounts	8	100%	Multi-bank balance aggregation
transactions.test.tsx	Ledger	14	100%	Canonical income/expense logging
transfer.test.tsx	Transfers	11	100%	Double-entry debit/credit match
budget.test.tsx	Budgeting	10	100%	Envelopes, 80% threshold alerts
intentEngine.test.ts	AI Engine	16	100%	11 regex intents, <50ms latency
supabaseSecurity.test.ts	Security RLS	15	100%	Cross-tenant query blocking
rpcExecution.test.ts	Atomic RPCs	12	100%	ACID rollback, bill payment
e2eJourney.test.tsx	Full E2E Flow	35	100%	End-to-end user lifecycle

4. Defect Log & Security Remediation (Rubric R2 - PO3)

DEF-01 (Critical): Vite bundled GEMINI_ keys into client JS. *Remediation:* Stripped prefix from Vite config, moved keys to Supabase Edge Functions. Verified in Git commit 053a9d0.
DEF-02 (High): Missing patterns property on new intent handlers (TS2741). *Remediation:* Added explicit regex arrays.
DEF-03 (Medium): Floating-point rounding. *Remediation:* Integer paise arithmetic round(amount * 100).
DEF-04 (Low): Missing account selector. *Remediation:* Integrated account dropdown with liquidity indicator.

5. User Acceptance Testing & Demonstration Script (Rubric R3 - PO4, PO9)

An 8-scenario UAT protocol covers onboarding, multi-bank tracking, expense logging, budget warnings, atomic transfers, bill settlement, AI queries, and security isolation. Status: *Pending External Action (Awaiting Panel Rehearsal)*.

A 15-minute demonstration script structure covers: (1) System overview, (2) Account setup, (3) Transaction logging & budget envelopes, (4) Atomic transfers, (5) Deterministic AI queries, and (6) Verification review. An offline sandbox fallback ensures demonstration resilience.

6. Research Paper Status & Academic Sign-Off (Rubrics R2/R3 - PO12, PSO1–PSO3)

The research paper has been revised with real architectural data and 184 passing tests. Publication status is explicitly documented as *Ready for Submission (Pending Faculty Advisor Sign-Off)*.

Academic Sign-Off: Aryan Kaundilya (Lead Architect), Devanshi Singh (QA Lead) | Expenzo Project Team | Sharda University